# My Notes

Coffee Shop Network Design

## Goal

Build a secure network for a small coffee shop that contains the following:

1) A reliable system to process customer payments
2) Free WIFI for customers

## Coffee Shop Structure

The Coffee shop Structure would be the following:

1) Manager office which contains the Manager PC + Office Printer
2) Checkout Counter which contains a POS Terminal + Receipt Printer
3) Guest-WIFI which is a Wireless Access Point
4) Small Network Cabinet which contains a Router + Multilayer Switch + ISP Connection to the router

**Note:** POS terminal is a device or a system that a business uses to accept payments and complete sales

## Devices used

- Cloud PT acting as the ISP / Internet
- 2911 Router
- 3560 Multilayer Switch
- 2 Printers & 2 PC
- Normal Access Point
- 2 Laptops acting as guests

## ISP Connection Method

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## Network Plan

| Area | VLAN | Subnet | Gateway | Switch ports | Addressing |
|---|---|---|---|---|---|
| Management Office | 10 | 192.168.10.0/24 | 192.168.10.1 | F0/1 - F0/5 | DHCP (.21+) & Printers: Static IP address |
| POS | 20 | 192.168.20.0/24 | 192.168.20.1 | F0/6 -F0/10 | DHCP (.21+0) & Printers: Static IP address |
| Guest-WIFI | 30 | 192.168.30.0/24 | 192.168.30.1 | F0/11 | DHCP (.21+) |
| Network Management | 99 | 192.168.99.0 | 192.168.99.1 | NONE (SVI is used) | Static IP address = 192.168.99.2 |

**Note:** The first 20 addresses in each subnet are reserved for gateways, pritners and future static devices

## Physical Connection

- Router G0/0 to Cloud Eth 6
- Router G0/1 to Switch G0/0
- Office PC to Switch F0/1
- Office Printer to Switch F0/2
- POS terminal to Switch F0/6
- Receipt Printer to Switch F0/7
- Access Point to Switch F0/11

**Note:** In Real life, you should known the following 3 things:

1) PC usually connect to the network through a cable connected to a wall jack, and this cable runs across the ceiling to the switch.
2) The Access Point gets power and network over one Ethernet cable using PoE
3) Its considered Best practice to leave extra switch interfaces for each department, and to keep network management on its own VLAN.

## Switch Configuration

### STEPS 1-5 (Basic Setup & Security)

1) Changed the hostname and disabled DNS lookup (`no ip domain-lookup`)
2) Set up a password and encrypted to access privileged exec mode Cisco1234
3) Encrypted the plain password in the running-config
4) Displayed a warning message (`banner motd *## Authorized USERS ONLY ##*`)
5) Set up a password for the console access password CCNA1234

### STEP 6 (Creating VLANS)

Create a VLAN for each department and assign them names

### STEP 7 (Assigning Ports)

**Assign Switch interfaces F0/1 - 5 to VLAN 10**
## Describe them as ## Management-Office Devices ##
enable Port-Fast on them

**Assign Switch interfaces F0/6 - 10 to VLAN 20**
## Describe them as ## POS Devices ##
enable Port-Fast on them

**Assign Switch interface F0/11 to VLAN 30**
## Describe them as ## Guest Wireless AP ##
enable Port-Fast on them

### STEP 8 (Trunk to the Router)

Configure the interface of the SW connected to the router as trunk port, give it a description, and allow all the VLANS that where created

### STEPS 9-10 (Configuring SVI & default-gateway)

assign VLAN 99 an IP address of 192.168.99.2 & Default gateway of 192.168.99.1

### STEP 11 (Configuring SSH for Remote Management)

- Set a switch domain (`ip domain-name coffeshop.local`)
- Create a local username & password (`username admin privilege 15 secret MO1234`)
- Generate a RSA key (`crypto key generate rsa`)
- Enable SSH V2 (`ip ssh version 2`)
- Enter the VTY lines (`line vty 0 15`)
- Allow SSH connections (`transport input ssh`)
- login local
- exit

### STEP 12 (Save the Current Configuration)

Save the running-config to the startup-config

## Router Configuration

### STEPS 1-5 (Basic Setup & Security)

1) Changed the hostname and disabled DNS lookup (`no ip domain-lookup`)
2) Set up a password and encrypted to access privileged exec mode Cisco1234
3) Encrypted the plain password in the running-config
4) Displayed a warning message (`banner motd *## Authorized USERS ONLY ##*`)
5) Set up a password for the console access password CCNA1234

### STEPS 6-7 (Router Interfaces Configuration)

1) Enable the Router interface facing the ISP , give it a description, and configure it to gets IP address from a DHCP server usually from the ISP
2) Enable the Router interface facing the Switch and give it a description

### STEP 8 (Router on a Stick Configuration)

Create the sub-interfaces for each VLAN and assign them the Gateway IP addresses

### STEP 9 (DHCP POOL Configuration)

1) Exclude the 1st 20 IP addresses from DHCP
2) Create the DHCP pool for each department and assign each DHCP pool its subnet, default gateway, and google DNS server which 8.8.8.8 for resolving domain names

### STEP 10 (Guest-WIFI ACL Configuration)

Configure an extended ACL on the guest VLAN's router sub-interface to block access to internal management,

- Assign the ACL GUEST-RESTRICTIONS
- Allow UDP DHCP client requests to any destination
- Block the Guest-WIFI traffic from the Management Office subnet
- Block the Guest-WIFI traffic from the POS network
- Block the Guest-WIFI traffic from Network Management network
- Allow all remaining IP traffic the Guest-WIFI to any destination
- Finally, apply the ACL inbound on the Guest-WIFI sub-interface

### STEP 11 (Save the Current Configuration)

Save the running-config to the startup-config

## End Devices Configuration

| Device | Configuration |
|---|---|
| Office Printer | Static 192.168.10.10 ; Gateway 192.168.10.1 ; DNS 8.8.8.8 |
| Receipt Printer | Static 192.168.20.10 ; Gateway 192.168.20.1 ; DNS 8.8.8.8 |
| Guest AP | WPA2-PSK ; SSID Guest-WIFI ; Key CoffeGuest2026 |